Consistent but Blind?

Image removal does not measure image reliance in medical vision-language models

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Paper, code, data and every number

The question

Medical VQA benchmarks score whether an answer is right, not whether the model *looked*. The *consistent-but-dangerous* taxonomy (Sadanandan & Behzadan, 2026) checks both, per answer:

	changes without the image	**unchanged** without the image
stable under paraphrase	IDEAL	DANGEROUS
flips under paraphrase	FRAGILE	WORST

Its reliance axis is **image removal**. A taxonomy like this invites comparisons: is the medical model more DANGEROUS than the general one?

We ask: does image removal rank models the way probes that **keep an image in the input** do?

Setup

Three pairs, each a medical model and the general model it shares an architecture with:

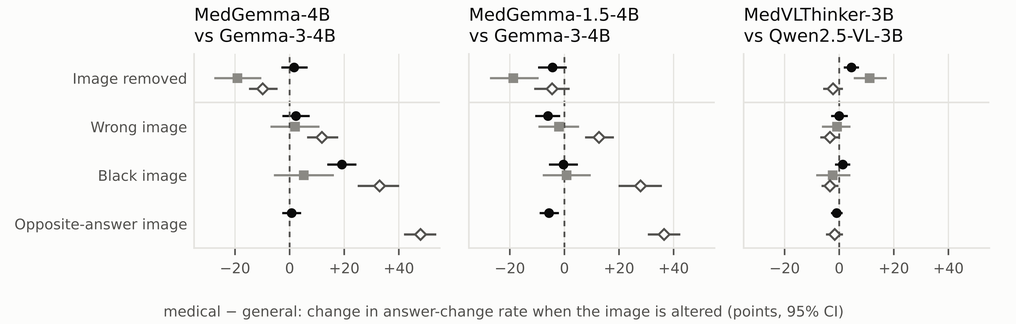
- **MedGemma-4B** and **MedGemma-1.5-4B** vs. **Gemma-3-4B**
- **MedVLThinker-3B** (RL on PMC-VQA) vs. its base **Qwen2.5-VL-3B**

Data: SLAKE (CT, MRI, X-ray), VQA-RAD (radiology), PathVQA (pathology): 3,365 questions on 907 images; 1,206 yes/no test questions are the primary set.

Probe	What changes
Image removed	image tokens omitted (the taxonomy's probe)
Wrong image	another image, same dataset and modality
Black image	a black image in the image slot
Opposite-answer image	another image on which the *same question* has the opposite answer
"Unknown" allowed	image removed, the model may decline

bf16, greedy, identical prompts; about 120,000 model calls on one 16 GB GPU. CIs: 95%, cluster bootstrap by image. **Pre-registered**: endpoints frozen before the main run; every later probe and model pair registered (a dated git tag) before it was run.

Finding 1. Image removal agrees with no in-format probe



Medical – general difference in how often a yes/no answer changes when the image is altered. A model that uses the image should change its answer when the image changes.

5 of 9
pair × dataset cells where image removal separates the models

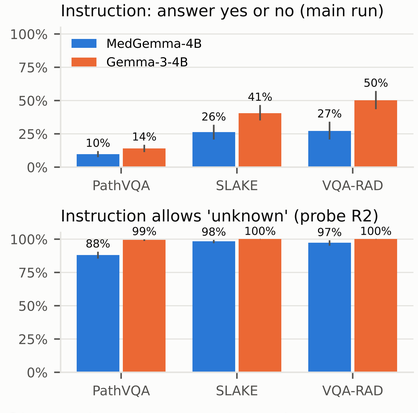
0 of 5
of those cells where any in-format probe agrees

+13.7
points more DANGEROUS for MedGemma-4B under image removal

What image removal picks up instead

Refusal. Told to answer yes or no about an image it was not given, **Gemma-3** would rather ask for it on **354 of 1,206** questions; **MedGemma-4B** on 219, **MedGemma-1.5** on 8, the Qwen pair never. The forced answers there are unrelated to the answers with the image ($\kappa \approx 0$), so about half count as "image-reliant". The largest image-removal gaps sit where refusal differs most (VQA-RAD).

Priors that match the image. A model whose blind guess matches what the image shows gives the same answer with and without it, and image removal calls that blind. MedGemma-4B's extra DANGEROUS answers are **all correct**: +14.1 correct, –0.4 wrong.



Share of yes/no questions the model declines without the image. Allowed to say "unknown" (bottom), both decline on ≥93%.

Finding 2. The in-format advantage stays in the training data

The same probe inside and outside the medical models' training data:

Follows the opposite-answer image, medical – general (points)	**SLAKE** (trained on)	**PathVQA** (held out)
MedGemma-4B – **Gemma-3**	+48.0	+0.8 [–2.7, 4.2]
MedGemma-1.5 – **Gemma-3**	+36.5	–5.6 [–9.1, –2.0]
MedVLThinker – **Qwen2.5-VL**	–1.6	–0.9 [–3.0, 1.3]

The PathVQA test was **registered before it was run**. Its MedGemma predictions failed, so the claim that they use the image more was **withdrawn**. All five models read pathology images to a similar degree (right on both images of a swap 36–43% of the time; a blind model: 0%).

The DANGEROUS share needs correcting

- **Chance floor:** 36–42% of yes/no answers look image-independent by chance alone.
- **Correctness:** the extra DANGEROUS answers are mostly *correct* answers.
- **No warning signal:** image reliance flags wrong answers at chance (AUROC **0.51**; confidence **0.71**).

Take-aways for reliance audits

1. Don't compare models with a DANGEROUS share measured by image removal: it tracks refusal and priors.
2. Probe in format, best with an image whose correct answer is the opposite.
3. Run that probe on data **outside** each model's training set.
4. Report the chance floor, split by correctness, and let models abstain.

Limits

Three pairs, two families: the problem exists; how common it is, we cannot say. The refusal and prior explanations are post hoc. R4 covers pathology only. Medical and general checkpoints differ in their whole post-training. Paraphrases were judged by an LLM; a 50-item check by the author, with AI assistance, accepted 88%. Benchmark VQA is not clinical use.